

Fake News Firewall: Designing a Scale-Free Social Network Engine for Misinformation Epidemic Simulation

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Abstract—This report describes the design, implementation, and analysis of the graph and network engine underlying *Fake News Firewall*, a group project that models misinformation propagation as a modified SEIR epidemic on a scale-free social network. The engine encompasses Barabási–Albert preferential attachment graph generation, Louvain community detection, betweenness centrality computation for bridge-node identification, and a force-directed layout algorithm. As the member responsible for this component, I discuss the technical decisions made at each stage, their mathematical grounding, the challenges encountered, and my personal analysis of how network topology—rather than content—governs misinformation dynamics. I further argue that the choice of graph model is not merely an implementation detail but the central scientific claim of the project.

Index Terms—Barabási–Albert model, scale-free networks, Louvain algorithm, betweenness centrality, epidemic spreading, misinformation, graph signal processing.

I. INTRODUCTION

MISINFORMATION on social platforms has been studied predominantly as a *content* problem: label it, remove it, or debunk it. Our project, *Fake News Firewall*, challenges this framing. We treat misinformation spread as a structural phenomenon—an epidemic process whose trajectory is determined by the topology of the underlying social network, not by the story itself.

The project models misinformation propagation using a modified SEIR (Susceptible–Exposed–Infected–Recovered–Vaccinated) epidemic model layered on a Barabási–Albert (BA) scale-free graph with Louvain community structure. An interactive browser-based simulator allows a player to deploy fact-checker tokens on network nodes, with the goal of keeping infection below 50% of the network within 20 ticks.

My contribution was building the **graph and network engine**: the backbone that every other component depends on. This report describes that work in detail, with particular emphasis on the mathematical properties of the graph model, the implications of those properties for epidemic dynamics, and my own analysis of the design choices made.

II. PROJECT BACKGROUND AND SHARED CONTEXT

A. Motivation

The Pizzagate conspiracy theory (2016) reached 1.7 million Facebook users before being debunked, resulting in a real-world armed incident at a Washington D.C. pizza restaurant

TABLE I
SYSTEM PARAMETERS

Symbol	Value	Description
N	120	Total nodes
K	4	Communities (Louvain)
m	2	BA attachment parameter
β	0.30	Base transmission rate
γ	0.18	Recovery rate per tick
δ	0.09	Relapse rate (post mutation)
$\text{Hub}_\times$	2.6	Hub amplification factor
T	5	Fact-checker token budget
Q	0.52	Louvain modularity

[5]. This case exemplifies a pattern documented in the literature: false news spreads faster and further than true news, not because humans are gullible, but because the network structure—the echo chambers, hubs, and bridges—amplifies transmission.

B. Group System Architecture

The project is divided into four components:

- 1) **Graph & Network Engine** (my contribution) — graph generation, community detection, centrality computation, layout.
- 2) **Epidemic Model** — modified SEIR state machine, probabilistic transitions, R_0 estimation.
- 3) **Frontend & Game UI** — HTML5 Canvas rendering, HUD, interaction.
- 4) **Analysis & Results** — strategy comparison experiments, figures, report coordination.

C. Model Parameters

Key parameters used across the system are summarised in Table I.

III. GRAPH AND NETWORK ENGINE DESIGN

A. Choice of Graph Model

The central scientific decision of the project was the choice of graph model. We required a model that satisfies three properties simultaneously:

- 1) **Scale-free degree distribution** — a small number of high-degree hub nodes with most nodes having low degree, consistent with empirical social networks [1].
- 2) **Community (echo chamber) structure** — dense intra-cluster connections with sparse inter-cluster bridges.
- 3) **Controllable bridge sparsity** — we needed to tune how rare cross-cluster edges are, since this determines whether the epidemic can jump between communities.

The Barabási–Albert preferential attachment model satisfies (1) by construction. We augmented it with a community-biased attachment rule to achieve (2) and (3). No existing standard model satisfies all three without modification.

B. Barabási–Albert Preferential Attachment

The BA model grows a graph by adding nodes one at a time. Each new node i attaches to m existing nodes. The probability of attaching to node j is:

$$P(\text{attach to } j) = \frac{k_j}{\sum_{\ell} k_{\ell}} \quad (1)$$

where k_j is the current degree of node j . This *preferential attachment* rule (“the rich get richer”) produces a power-law degree distribution:

$$P(k) \sim k^{-\gamma}, \quad \gamma \approx 3 \quad (2)$$

This means that while most nodes have degree 2–4, a few hub nodes accumulate degrees of 30 or more—mirroring the follower distribution on real platforms such as Twitter.

1) *Community-Biased Attachment*: Standard BA generates a single globally connected component without community structure. I extended it with a *cluster bias*: when selecting which existing nodes to attach to, each new node i belonging to cluster c_i uses a biased sampling strategy:

$$P(\text{pool} = \text{same cluster}) = 0.72 \quad (3)$$

With 25% probability the pool is all nodes regardless of cluster. Within either pool, equation (1) still governs selection. This produces dense intra-cluster edges and sparse cross-cluster bridges naturally, without hard-coding community assignments.

C. Hub Identification

After graph construction, hub nodes are identified as those whose degree exceeds the 92nd percentile across all nodes. Hubs receive a transmission amplification of $2.6\times$ in the epidemic model, simulating algorithmic amplification on social media platforms.

D. Community Detection via Louvain

To identify the community structure after graph generation—and to validate that the biased attachment procedure produced genuine clusters—I implemented the Louvain modularity maximisation algorithm [2].

The algorithm maximises modularity:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (4)$$

where A_{ij} is the adjacency matrix, k_i, k_j are node degrees, m is the total number of edges, and $\delta(c_i, c_j) = 1$ if nodes i and j belong to the same community.

On our generated graphs, Louvain consistently yields $Q \approx 0.52$, which confirms strong community separation. The algorithm runs in two phases:

- 1) **Local movement**: each node is moved to the community of its neighbour that maximally increases Q .
- 2) **Aggregation**: communities are compressed into super-nodes and the procedure repeats until Q no longer improves.

The community labels produced by Louvain are used by the frontend to assign node colours (one per community) and to identify bridge edges as those connecting nodes of different community labels.

E. Bridge Node Detection via Betweenness Centrality

A key insight of the project is that the *structural* importance of a node for misinformation containment is not captured by degree alone. A bridge node with degree 2 connecting two clusters may be far more critical than a hub with degree 40 entirely inside one cluster.

Betweenness centrality captures this:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma(s, t | v)}{\sigma(s, t)} \quad (5)$$

where $\sigma(s, t)$ is the number of shortest paths from s to t , and $\sigma(s, t | v)$ is the number of those paths passing through v .

I computed betweenness centrality using Brandes’ algorithm [3], which runs in $O(VE)$ time for unweighted graphs. For our $N = 120$ node graph this is negligible. The resulting scores are exported alongside the adjacency list for use by other team members.

1) *Degree vs. Betweenness: A Concrete Example*: Consider a bridge node b with degree 2 connecting cluster A (60 nodes) to cluster B (60 nodes). Every shortest path from any node in A to any node in B passes through b . Its betweenness score is therefore:

$$C_B(b) = \frac{60 \times 60}{N(N-1)/2} \approx 0.61 \quad (6)$$

This places b in the top 5% by betweenness, despite being the 80th percentile by degree. Our results confirm: vaccinating high-betweenness nodes achieves an 84% win rate versus 31% for degree-centrality targeting [8].

F. Force-Directed Layout

The graph layout algorithm must separate clusters visually so players can identify community boundaries and bridge edges during gameplay. I implemented a modified Fruchterman–Reingold force-directed algorithm [4] with 140 iterations, augmented by an additional *cluster-pull* term:

$$\mathbf{f}_{\text{pull}}(v) = \alpha \cdot (\mathbf{c}_{c_v} - \mathbf{x}_v) \quad (7)$$

where $\mathbf{c}_{c_v}$ is the centroid of node v ’s community and $\alpha = 0.045 \cdot \text{cool}(t)$ is a temperature-decaying pull coefficient.

Without this term, the standard algorithm fails to separate communities visually even when modularity is high, because it has no knowledge of community membership.

The result is a layout where echo chambers are visually distinct elliptical clusters, bridge edges appear as long amber-coloured connections between clusters, and hub nodes are the largest circles within each cluster.

G. Data Structure and API

The engine exports:

- `nodes[]` — array of node objects with position, state, degree, cluster, isHub flag.
- `edges[]` — array of $\{a, b\}$ pairs.
- `adj[]` — adjacency list for $O(1)$ neighbour lookup during epidemic simulation.

This clean separation allowed the epidemic model, frontend, and analysis components to be developed in parallel without depending on implementation details of the graph engine.

IV. INDIVIDUAL ANALYSIS AND PERSONAL INSIGHTS

A. Why the Graph Model Is the Central Scientific Claim

During development I came to understand that the choice of graph model is not merely an engineering decision — it *is* the hypothesis. If we had used an Erdős–Rényi random graph with uniform degree distribution, betweenness centrality would not systematically outperform degree centrality, because there would be no meaningful bridge nodes. The scale-free, modular topology is precisely what makes the intervention strategy finding non-trivial.

This realisation changed how I thought about the project. The graph engine is not “the part that makes the other parts work.” It is the part that determines whether the scientific result is real or an artifact of the model choice. I spent significant time validating that the biased BA procedure actually produces the intended structure, using modularity Q as the validation metric and running the Louvain detection post-hoc as a sanity check rather than just for colouring.

B. The Stochastic Extinction Problem

One surprising discovery was that games can be *won without any player intervention*. The epidemic starts from a single Patient Zero node. With base transmission $\beta = 0.30$ and recovery $\gamma = 0.18$, there is a non-negligible probability that Patient Zero recovers before infecting any neighbour. From SIR branching process theory, the probability of early extinction from a single seed is approximately:

$$P(\text{extinction}) \approx \left(\frac{\gamma}{\beta}\right)^1 = \frac{0.18}{0.30} = 0.60 \quad (8)$$

This means roughly 40–60% of games starting from a low-degree Patient Zero will self-extinguish — a statistically valid outcome, not a bug. This has an important implication for the intervention strategy experiments: results must be averaged over at least 100 runs per strategy to wash out this variance, which is exactly the methodology we adopted.

From a graph perspective, the extinction probability is also modulated by the degree of Patient Zero. A seed in a high-degree hub has more transmission opportunities per tick, making early extinction less likely. This means the *location* of Patient Zero on the graph, not just the epidemic parameters, determines game difficulty. A fuller investigation would condition results on Patient Zero degree.

C. Limitations of the Graph Model

1) *Static Topology*: Our graph is generated once and remains fixed throughout the simulation. Real social networks are dynamic: users follow and unfollow, platform algorithms rewire effective edge weights by amplifying engaging content. An engagement-driven feed can be modelled as re-weighting edges toward Infected (actively sharing) nodes each tick. I believe this would raise effective R_0 during the early outbreak phase and make bridge vaccination even more critical, but this remains to be tested.

2) *Unweighted Edges*: We treat all edges as equal. In practice, a retweet from a 5-million-follower account reaches vastly more people than one from a regular user. A weighted version of the graph, where edge weight encodes interaction frequency, would require re-deriving all centrality measures and potentially change the relative ranking of degree vs. betweenness as intervention targets.

3) *Synthetic Validation Only*: All results are validated on graphs generated by our own procedure. We have not tested on empirical social network data (e.g. a real Twitter follow graph). The finding that betweenness centrality outperforms degree centrality may not generalise to graphs with different modularity or degree distribution properties.

D. Lessons Learned

Force layout requires domain knowledge. The standard Fruchterman–Reingold algorithm does not know about communities. Without the cluster-pull term I added, the layout produced visually tangled graphs where players could not distinguish clusters. The fix was not a bug in the algorithm but a missing domain-specific term. This taught me that applying a general algorithm to a domain-specific problem almost always requires domain-aware modification.

Modularity can be maximised trivially. Louvain can sometimes assign all nodes to one community (maximising Q at zero by trivially satisfying $\delta(c_i, c_j) = 1$ everywhere) when the graph is too dense. I had to tune the inter-cluster edge probability to 0.08 to prevent this degenerate case while still producing visible bridges.

Betweenness is the right metric but expensive to explain. Computing betweenness centrality is conceptually straightforward but explaining *why* a degree-2 node outranks a degree-40 node requires the audience to internalise the notion of shortest paths as information channels. This was the hardest part of preparing the project presentation — not the code, but the intuition transfer.

E. Connection to Graph Signal Processing

From the perspective of graph signal processing (GSP), misinformation spread can be interpreted as a graph signal $\mathbf{x}(t)$ on node space, where $x_i(t) \in \{0, 1\}$ indicates whether node i believes the story at tick t . The epidemic propagation operator is then a non-linear function of the graph Laplacian: high-frequency components (localised signals in one cluster) are preserved by the community structure, while low-frequency components (global consensus) only emerge when bridge edges carry signal between clusters.

Vaccination can therefore be interpreted as *graph filtering*: removing a bridge edge zeroes out the cross-cluster component of the propagation operator, preventing low-frequency signal spread. This framing connects our project directly to the course material on graph Fourier transforms and spectral graph theory, though we did not implement a formal spectral analysis.

V. CONCLUSION

I designed and implemented the graph and network engine for Fake News Firewall: a Barabási–Albert scale-free graph with community-biased attachment, Louvain community detection ($Q \approx 0.52$), betweenness centrality computation for bridge identification, and a modified force-directed layout for visual separation of echo chambers.

My principal insight is that the graph model is not infrastructure — it is the hypothesis. The finding that betweenness centrality outperforms degree centrality by $2.7\times$ in win rate is only meaningful because the graph has sparse, structurally critical bridge nodes. A different graph model would produce a different result.

The project taught me that network science tools from this course — centrality, modularity, degree distribution, preferential attachment — apply directly to one of the most consequential policy problems of our time. Misinformation is a topology problem. Solving it requires understanding the structure of the network, not just the content of the story.

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